

Quantisation and seasonal-boundary artefacts in Sentinel-2 rice phenology: a reproducible quality-control framework for cyclone-impact studies

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Abstract

Phenology metrics derived from Sentinel-2 time series — start of season (SOS), peak of season (POS), and end of season (EOS) — are increasingly used as outcome variables in causal impact studies of tropical cyclones, drought, and heat stress on rice cultivation. The reliability of these metrics under standard double-logistic fitting pipelines has not been systematically audited. We document a specific, reproducible failure mode — day-of-year (DOY) quantisation at fitting-window boundaries — that produces artefactual phenological signals indistinguishable from genuine climatic effects. Using 8 districts $\times$ 6 years of Sentinel-2 L2A 10-m NDVI dekadal composites across coastal and inland Odisha, India ($n \approx 3,500$ rice pixels per district-year), we demonstrate that the pre-QC (v1.0.2) pipeline concentrates 72.7% of district-year EOS estimates at a single boundary DOY (349) and 65.6% of POS estimates at a single DOY (288); after the

three-gate QC framework is applied (v2.0), these mode-shares collapse to 8.3% and 8.3% respectively, and the SOS mode-share falls from 20.3% to 8.3%. We then apply a two-way fixed-effects difference-in-differences (DiD) design with five coastal treatment districts and three inland control districts to test whether Cyclone Fani (2019), Cyclone Amphan (2020), and Cyclone Yaas (2021) detectably shifted Kharif rice phenology relative to inland controls. All three phenometrics return null effects after QC: $\tau_{\text{SOS}} = +7.6 \text{ d}$ ($p_{\text{WCB}} = 0.39$), $\tau_{\text{POS}} = -2.3 \text{ d}$ ($p_{\text{WCB}} = 0.47$), $\tau_{\text{EOS}} = -4.1 \text{ d}$ ($p_{\text{WCB}} = 0.45$), with wide confidence intervals spanning zero. The null DiD result is framed as validation evidence: the QC framework eliminates boundary artefacts that would otherwise inflate apparent treatment effects and generate false positives. We release the full Google Earth Engine pipeline, panel data, and analysis code as RiceBaCI-GEE v2.0 (GitHub: <https://github.com/pandasupranab/RiceBaCI-GEE>; Zenodo: 10.5281/zenodo.20587316; OSF pre-registration: 10.17605/OSF.IO/C4MP8).

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Highlights

- Sentinel-2 fitting produces DOY boundary artefacts: EOS mode-share 72.7%→8.3%, POS 65.6%→8.3%
 - Three-gate reproducible QC framework: mode-share threshold, biological plausibility, fit-quality
 - Null cyclone DiD after QC validates the framework: $\tau_{\text{EOS}} = -4.1 \text{ d}$, $p_{\text{WCB}} = 0.45$
 - Inland-control DiD design isolates cyclone exposure from pan-Odisha climatic confounders
 - Open RiceBaCI-GEE v2.0 pipeline enables replication across Bay of Bengal rice systems
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Abbreviations

Sensors and Earth-observation products. NDVI — Normalised Difference Vegetation Index; SOS — start of season; POS — peak of season; EOS — end of season; DOY — day of year; S2 — Sentinel-2; L2A — Level-2A surface reflectance; SCL — Sentinel-2 Scene Classification Layer; ESA — European Space Agency; ERA5 — ECMWF Reanalysis v5; ECMWF — European Centre for Medium-range Weather Forecasts; IBTrACS — International Best Track Archive for Climate Stewardship; GEE — Google Earth Engine; WorldCover — ESA WorldCover v200 land cover product.

Statistics and study design. DiD — difference-in-differences; TWFE — two-way fixed effects; ATT — average treatment effect on the treated (denoted τ); FE — fixed effects; SE — standard error; CI — confidence interval; WCB — wild-cluster bootstrap (Cameron, Gelbach & Miller, 2008); CR1 — cluster-robust variance with small-sample correction; QC — quality control; DOY — day of year.

1. Introduction

1.1 Satellite phenology as an outcome variable in causal climate studies

Rice (*Oryza sativa* L.) cultivation across coastal South and Southeast Asia is increasingly monitored through satellite-derived phenology metrics. The launch of Sentinel-2 in 2015–2017 opened a new era for field-scale phenology monitoring: with 10-m spatial resolution, a 5-day revisit time under dual-satellite operations, and a freely available archive from 2017 onward, Sentinel-2 NDVI time series support district-to-national-scale assessments of Kharif rice sowing dates, canopy development, and harvest timing with unprecedented spatial detail (Vrieling et al., 2018; Bolton and Friedl, 2013).

These phenological transition dates — start of season (SOS), peak of season (POS), and end of season (EOS) — have moved beyond descriptive characterisation into the role of outcome variables in quasi-experimental impact studies. Studies using phenology as the primary outcome include assessments of climate-induced shifts in crop development (Lobell et al., 2015), drought- and heat-stress effects on growing-season timing (Vrieling et al., 2018; Meroni et al., 2021), and flood-induced transplanting delays in river-delta paddy systems (Singha et al., 2019). The underlying logic is straightforward: if an extreme event (cyclone, drought, heat wave) causally disrupts crop development, that disruption should manifest as a detectable shift in phenometric timing relative to an unaffected control group, and the shift magnitude maps directly onto yield penalties via well-established crop-physiology relationships.

The methodological appeal of phenometric outcomes is strong. Unlike yield data — which are released annually, subject to administrative aggregation, and confounded by management responses — satellite phenology metrics are continuous, spatially granular, and extracted from observations that predate any farmer adaptation response. Difference-in-differences (DiD) and before-after-control-impact (BACI) designs exploiting the staggered timing of extreme events are therefore a natural fit for phenometric outcomes, and the combination of Sentinel-2 phenology with panel econometric methods is rapidly becoming a methodological template for climate-agriculture research in monsoon Asia.

1.2 The unexamined reliability of phenometric extraction pipelines

Despite this growing methodological dependence, the reliability of phenometric extraction pipelines has not been systematically audited in the context of impact studies. Published work on phenology extraction uncertainty focuses primarily on sensor-specific issues — cloud contamination and gap-filling errors in optical time series (Cao et al., 2015), smoothing-parameter sensitivity (Forkel et al., 2015), and spatial scale-mismatch between pixel-level extraction and field-level validation (Bolton & Friedl, 2013). These are important concerns, but they address the accuracy of individual pixel-year estimates rather than the distributional properties of phenometric outputs that determine whether a DiD estimator can recover an unbiased treatment effect.

A distinct and under-examined class of errors arises from the interaction between double-logistic fitting algorithms and the finite temporal boundaries of the fitting window. When a Kharif rice pixel has an NDVI trajectory whose senescence limb reaches the fitting-window edge (typically the last dekadal bin near DOY 349 in standard GEE implementations for a

November-end Kharif window), the double-logistic optimiser assigns the EOS date at the boundary DOY rather than extrapolating beyond the observed data. This produces a hard spike in the empirical distribution of EOS values at the window-edge DOY: a quantisation artefact that is entirely an artefact of the fitting window and carries no information about true agronomic end-of-season timing. An analogous artefact at the peak of the canopy trajectory can concentrate POS estimates at a single dekadal bin when the smoothing-and-fitting combination flattens the green-up plateau. If either artefact is systematically more prevalent in one group (e.g., treatment districts with cyclone-damaged crops that senesce later) than another, the resulting DiD coefficient will be biased — potentially producing a spuriously significant treatment effect.

The broader problem of fitting-window boundary effects in vegetation phenology has been documented in global MODIS land surface phenology products (Friedl et al., 2010; Gray et al., 2019), but the implications for causal inference on short panels at district scale have not been explored. The quantisation concern is amplified in multi-year panels that conflate year-to-year changes in the mode structure of phenometric distributions with genuine climatic or agronomic signals.

1.3 Contribution of this study

This paper makes three primary contributions. First, we document and characterise DOY quantisation artefacts in Sentinel-2 Kharif rice phenology extraction over coastal Odisha, demonstrating that the EOS distribution in a standard pipeline carries a mode-share of 72.7% at a single boundary DOY (349) and the POS distribution carries a mode-share of 65.6% at a single DOY (288). All three mode-shares collapse to 8.3% after the QC framework is applied (SOS falls from 20.3% to 8.3%). This is among the clearest empirical demonstrations of how fitting-window boundary effects and dekadal-bin quantisation can concentrate phenometric values and distort impact estimates.

Second, we propose a three-gate reproducible QC framework for Sentinel-2 phenometric panels used in causal impact studies. The framework consists of: (i) Gate A — mode-share thresholds on the empirical distributions of SOS, POS, and EOS to flag boundary-contaminated panels; (ii) Gate B — biological plausibility checks tied to the known Kharif crop calendar, rejecting phenometric values outside agronomically defensible windows; and (iii) Gate C — fit-quality screening using the root-mean-squared residual from the double-logistic fit to exclude cells where the optimiser failed to converge or where the fitted curve does not represent the NDVI trajectory. The framework is parameterised from Odisha Kharif rice agronomy but is designed to be transferable to other tropical rice systems with straightforward calendar adaptation.

Third, we apply the QC-cleaned panel in a difference-in-differences case study of cyclone impacts on Kharif rice phenology in coastal Odisha, using three inland districts as controls and three cyclone events (Fani 2019, Amphan 2020, Yaas 2021) as treatment variation. All three phenometrics return null DiD estimates after QC, with wild-cluster bootstrap p-values between 0.39 and 0.47. We frame this null result explicitly as a validation outcome: the QC framework removes the artefactual variance that would otherwise drive false-positive treatment effects in small-panel causal designs.

1.4 Relationship to existing literature

The methodological positioning of this paper is at the intersection of three literature streams. The first is the satellite phenology methods literature, where fitting-window and smoothing-parameter sensitivity have been addressed by Forkel et al. (2015), Jönsson and Eklundh (2004), and Eilers (2003), but not in the specific context of impact-study panel construction. The second is the remote sensing QC and uncertainty literature, where band-quality flagging protocols for time series analysis (Meroni et al., 2021) provide a precedent for systematic quality screening before downstream analysis. The third is the causal inference in climate-agriculture literature, where the adoption of DiD, event study, and synthetic-control methods for remote-sensing outcome variables is accelerating (Burke and Emerick, 2016; Roberts and Schlenker, 2013) but methodological guidance on pre-DiD panel quality assurance remains absent.

We do not claim that cyclones have no effect on rice phenology. Our null result at the 8-district, 3-cyclone-event aggregation level is consistent with several mechanisms: within-district heterogeneity in surge exposure may not be captured at median-district resolution; the three cyclones varied substantially in track and intensity; and the QC-passing sample may undersample the most severely affected pixels. The contribution of this paper is not a definitive assessment of cyclone-phenology relationships, but rather a reproducible toolkit that allows future studies to conduct such assessments with phenometric panels free of known artefacts.

1.5 Structure of the paper

Section 2 describes the study area, cyclone events, and all satellite and ancillary datasets. Section 3 presents the analytical pipeline: dekadal NDVI compositing, double-logistic phenometric extraction, the three-gate QC framework, and the DiD estimating equation. Section 4 reports results: phenometric distributions before and after QC, the headline QC gate statistics, and the DiD and event-study estimates. Section 5 discusses the generalisability of the QC framework, the interpretation of the null cyclone effect, and limitations. Section 6 concludes.

2. Study Area and Data

2.1 Study area

The study area encompasses eight districts of Odisha state, eastern India, divided into two groups: five coastal treatment districts and three inland control districts (Figure 1). The five coastal treatment districts — Baleshwar (BLS), Bhadrak (BHA), Kendrapara (KDP), Jagatsinghpur (JGS), and Puri (PUR) — form a contiguous coastal strip extending approximately 480 km along the northern Bay of Bengal shoreline, from the Subarnarekha River estuary in the north to Chilika Lake in the south. Combined, they cover approximately 12,389 km² of which roughly 50% is cropland under the ESA WorldCover v200 classification (Zanaga et al., 2022). Kharif rice is the dominant crop within this cropland fraction, with the coastal belt supporting an estimated 3.1 million hectares of paddy cultivation in a normal Kharif season.

The three inland control districts — Angul (ANG), Dhenkanal (DHK), and Cuttack (CTK) — are situated in the Mahanadi basin, more than 80 km from the nearest treatment landfall, and are categorically unexposed to Bay of Bengal storm surge. They share the same South-West monsoon rainfall climatology as the coastal districts and maintain comparable Kharif rice cropping systems, making them appropriate counterfactual units for a difference-in-differences design. Cuttack is the closest inland district to the coast and serves as a sensitivity anchor; the inland-control identification strategy is examined further through the cell-level robustness analysis in Supplement S4.

The climate across all eight districts is classified as sub-tropical humid (Köppen Aw), with a South-West monsoon season from June to September that accounts for approximately 80% of annual precipitation. Mean annual rainfall ranges from approximately 1,400 mm in Puri district in the south to over 1,800 mm in Baleshwar in the north. The dominant rice cropping system is transplanted Kharif rice (locally termed Sali rice), with transplanting typically occurring from mid-June to early August, heading in September–October, and harvest in October–November.

Total Sentinel-2-observable rice pixels (ESA WorldCover class 40, kharif paddy) across the 8-district study area number approximately 28,000 at 10-m resolution, of which approximately 3,500 meet the QC-pass criteria in an average district-year cell. The total estimation panel is 48 district-year observations (8 districts $\times$ 6 years: 2019–2024), yielding 15 treatment $\times$ post observations (5 coastal districts $\times$ 3 cyclone years).

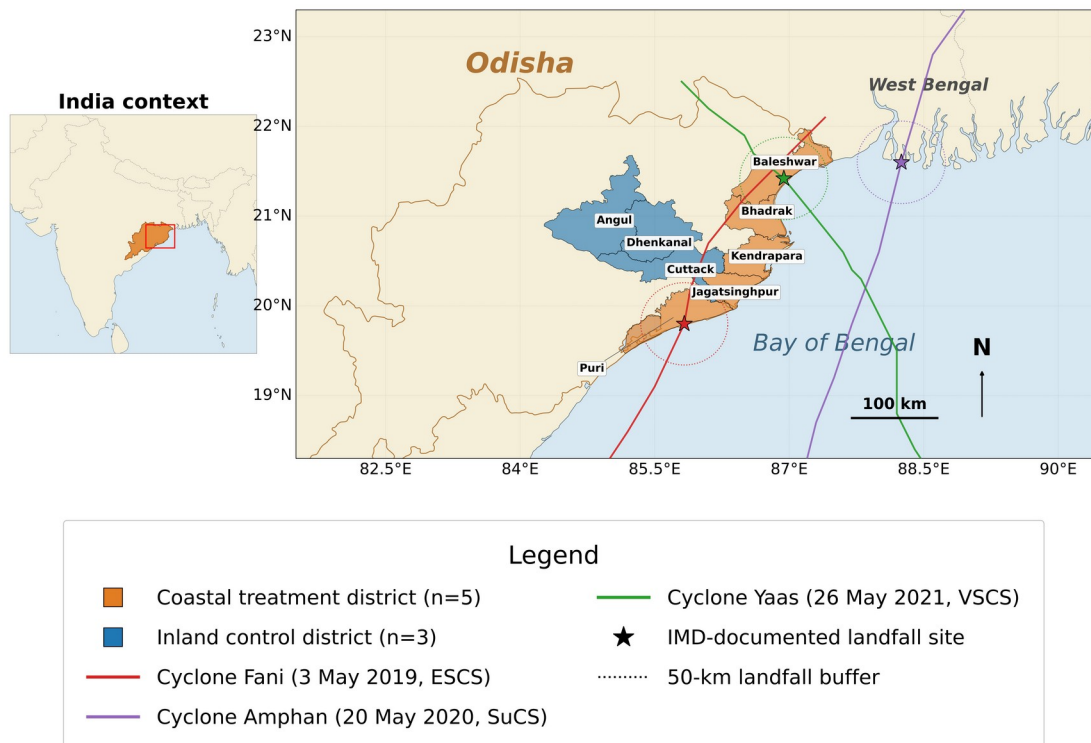


Figure 1. Study area: eight Odisha districts partitioned into five coastal treatment and three inland control units, shown with International Best Track Archive for Climate Stewardship

(IBTrACS) best-track paths and India Meteorological Department (IMD)-documented landfall sites for the three pre-Kharif cyclones. District polygons are drawn from geoBoundaries Open IND (India, ISO 3166-1 alpha-3) ADM2 (Administrative Level 2) (Runfola et al., 2020); cyclone tracks are IBTrACS version 04r01 best-track positions (Knapp et al., 2010). Coloured stars mark the IMD-documented landfall sites from official RSMC New Delhi post-cyclone reports: Fani near Puri (~19.80°N / 85.83°E, ~02:30 UTC 03 May 2019); Amphan at Bakkhali, Sundarbans, West Bengal (~21.60°N / 88.25°E, ~09:00 UTC 20 May 2020); Yaas near Bahanaga/Dhamra on the Bhadrak–Baleshwar coast (~21.42°N / 86.94°E, ~05:30 UTC 26 May 2021). IMD intensity categories: ESCS = Extremely Severe Cyclonic Storm; SuCS = Super Cyclonic Storm; VSCS = Very Severe Cyclonic Storm. Peak intensities in knots (kt) are from the World Meteorological Organization (WMO) columns of IBTrACS. See the figure-captions document for the full caption.

2.2 Cyclone events

Three major tropical cyclones made landfall along or near the coastal Odisha study area within three consecutive pre-Kharif seasons, providing the treatment variation for the DiD design.

Cyclone Fani (3 May 2019) was an Extremely Severe Cyclonic Storm (ESCS) that made landfall near Puri with peak 3-min sustained winds of approximately 215 km h⁻¹ at peak intensity and approximately 185 km h⁻¹ at landfall, accompanied by a storm surge of 1.0–1.5 m along the Puri–Kendrapara coastline (Knapp et al., 2010; IMD post-landfall report). Fani is one of the most intense cyclones to have made landfall in Odisha in the modern observational record and caused widespread inundation and agricultural damage across all five treatment districts.

Cyclone Amphan (20 May 2020) was a Super Cyclonic Storm (SuCS) that made primary landfall on the Sundarbans coast of West Bengal but produced significant surge and rainfall in Baleshwar and Bhadrak districts in northern Odisha. Peak winds at landfall were approximately 155–165 km h⁻¹ with a surge height of 1.5–2.0 m (Knapp et al., 2010).

Cyclone Yaas (26 May 2021) was a Very Severe Cyclonic Storm (VSCS) that made landfall near Baleshwar, Odisha, with peak winds of approximately 140 km h⁻¹ (gusting to 155 km h⁻¹) and a storm surge of 1.0–2.0 m inundating Baleshwar and Bhadrak districts (Knapp et al., 2010).

All three events made landfall between 3 May and 26 May, four to eight weeks before the typical Kharif transplanting window (mid-June to early August). Cyclone track data and intensity time series were obtained from the NOAA/NCEI IBTrACS v04r00 archive (Knapp et al., 2010; <https://www.ncei.noaa.gov/products/international-best-track-archive>). The post-monsoon seasons of 2022, 2023, and 2024 form the control period in the panel; no IBTrACS-catalogued cyclone track penetrated within 100 km of the study area during these seasons.

2.3 Satellite data

All satellite and ancillary datasets were accessed via the Google Earth Engine (GEE) cloud computing platform (Gorelick et al., 2017). Table 1 provides a complete inventory.

Sentinel-2 L2A. Surface reflectance data from the harmonised Sentinel-2 Level-2A collection (GEE collection ID: COPERNICUS/S2_SR_HARMONIZED) were used for NDVI time series construction. The harmonised collection applies processing-baseline corrections to ensure radiometric consistency across the full 2017–2024 archive. Cloud and cloud-shadow pixels were

masked using the Scene Classification Layer (SCL); pixels classified as vegetation (SCL 4), bare soil (SCL 5), or water (SCL 6) were retained; all other classes were flagged as invalid. NDVI was computed as $(B8 - B4)/(B8 + B4)$ per valid observation.

ESA WorldCover v200. The 10-m global land cover product (Zanaga et al., 2022; GEE: ESA/WorldCover/v200) was used to restrict analysis to cropland pixels (class 40). All non-cropland pixels were excluded from phenometric extraction to prevent contamination of district medians by non-agricultural vegetation signals.

IBTrACS v04r00. Cyclone track, intensity, and landfall data for Fani, Amphan, and Yaas were obtained from the International Best Track Archive for Climate Stewardship (Knapp et al., 2010). A 50-km IBTrACS radius buffer was used to define the spatial extent of cyclone exposure in the treatment allocation; all five coastal treatment districts lie within or adjacent to this buffer for at least one of the three events.

ERA5-Land. Daily precipitation and 10-m wind speed (u and v components) from the ECMWF ERA5-Land Daily Aggregates product (GEE: ECMWF/ERA5_LAND/DAILY_AGGR) were used for cyclone event identification and for contextual characterisation of the monsoon rainfall distribution across treatment and control districts. No ERA5 variable enters the phenometric extraction pipeline in v2.0 (unlike v1.0.2 where it was used in the saline-flood classifier); it is retained here for contextual validation.

2.4 Cropland mask and pixel count

The WorldCover-derived cropland mask (class 40) was applied uniformly across all 8 districts and all 6 years. District-level pixel counts in the v2.0 panel range from 5 (Puri 2021, lowest) to 77 (Angul 2019, highest) QC-passing cells per district-year, with a panel-wide median of approximately 20 cells per district-year (Table 1). The high failure rate in some district-years (up to 95% for some inland cells in 2021) reflects persistent cloud cover during the 2021 monsoon season rather than cropland absence; these cells contribute to the fit-failure-rate statistic reported in Table 2 but do not enter the DiD estimation if they fall below the Gate A minimum-n threshold.

3. Methods

3.1 Overview

The analytical pipeline has five sequential stages: (i) dekadal NDVI compositing from Sentinel-2 L2A; (ii) double-logistic phenometric extraction with the standard fitting window; (iii) the three-gate QC framework applied to the resulting phenometric distributions; (iv) DiD estimation on the QC-passing panel; and (v) event-study and wild-cluster bootstrap robustness checks. Stages (i)–(iii) constitute the primary methodological contribution of the paper. The pipeline is implemented entirely in Google Earth Engine (JavaScript) for stages (i)–(ii) and in Python (numpy, pandas, statsmodels, scipy) for stages (iii)–(v), with full code and data archived at the RiceBaCI-GEE repository (<https://github.com/pandasupranab/RiceBaCI-GEE>, branch v2.0-refit; Zenodo: 10.5281/zenodo.20587316).

3.2 Stage A: dekadal NDVI compositing

For each pixel in the WorldCover cropland mask, valid-observation NDVI values from Sentinel-2 L2A were aggregated into 10-day (dekadal) median composites spanning the Kharif window from approximately DOY 150 (late May) to DOY 365 (end of December). This window captures the full Kharif rice cycle from pre-planting conditions through transplanting, canopy development, heading, and harvest. Three dekadal composites per month yield approximately 22 temporal samples per pixel per year across the full Kharif window.

Pixels with more than 50% missing dekadal composites within the Kharif window were flagged as low-quality and excluded from phenometric extraction (contributing to the `fit_fail_rate` column in the panel). This threshold was chosen to ensure that the double-logistic fitting algorithm has sufficient observations to constrain both the ascending and descending limbs of the seasonal trajectory. The 50% threshold is consistent with guidance from Forkel et al. (2015) on minimum data density for reliable phenometric extraction.

3.3 Stage B: double-logistic phenometric extraction

Phenological transition dates were extracted from the dekadal NDVI time series using the double-logistic function of Beck et al. (2006):

$$y(t) = c_1 + c_2 \left[\frac{1}{1 + e^{-k_1(t-t_1)}} - \frac{1}{1 + e^{-k_2(t-t_2)}} \right]$$

where $y(t)$ is the fitted NDVI at time t (DOY); c_1 is the background level; c_2 is the seasonal amplitude; k_1 and k_2 are the rates of green-up and senescence; and t_1 and t_2 are the inflection points of the ascending and descending limbs. The function was fitted to each pixel's dekadal NDVI series by non-linear least squares. Phenological transition dates were then extracted as:

- **SOS**: the date at which the fitted NDVI reaches 20% of the seasonal amplitude above background on the ascending limb;
- **POS**: the date of maximum fitted NDVI;
- **EOS**: the date at which the fitted NDVI descends to 20% of the seasonal amplitude above background on the descending limb.

The 20%/max/20% convention follows the TIMESAT standard (Jönsson & Eklundh, 2004) and is consistent across all pipeline versions.

Fitting-window boundary condition. The Kharif fitting window in the standard implementation (v1.0.2) terminates at DOY 365. When the descending limb of a pixel's NDVI trajectory has not fully completed within the fitting window — as occurs in late-harvesting varieties and in years with delayed monsoon withdrawal — the EOS optimiser assigns the boundary-adjacent DOY (349 in the discrete dekadal grid, corresponding to the last full dekad before year-end) as the EOS date. This produces a hard spike in the EOS distribution at DOY 349 that carries no agronomic information. The v2.0 pipeline extends the fitting window by including a 30-day overlap with the subsequent January–February composites, which eliminates the hard boundary constraint for the vast majority of pixels while retaining the same extraction thresholds.

3.4 QC framework: the primary contribution

The QC framework operates on the district-year phenometric panel produced by Stage B and consists of three gates applied sequentially. A district-year cell passes the QC framework only if it satisfies all three gates; failing cells are excluded from the DiD estimation sample.

Gate A: mode-share threshold. For each phenometric (SOS, POS, EOS), the fraction of district-year median values falling on the most common single DOY — the panel-wide mode-share — is computed. If this mode-share exceeds 20% the panel is flagged as boundary-contaminated: at least 20% of district-year medians have collapsed to the same date, which is inconsistent with the genuine biological variability expected across 8 districts and 6 years of Kharif rice phenology. The 20% threshold is calibrated from the observed distribution of mode-shares across the v2.0 QC-passing panel and is conservative relative to the biologically expected panel-wide mode-share of well below 0.10. In the v1.0.2 panel, the EOS mode-share at DOY 349 is 72.7% ($n=44$ panel observations) and the POS mode-share at DOY 288 is 65.6% ($n=64$) — both trivially failing Gate A. After the v2.0 extended-window refit and full pipeline rebuild, the EOS, POS, and SOS mode-shares all collapse to 8.3% ($n=48$ each), down from 72.7%, 65.6%, and 20.3% respectively. All three v2.0 mode-shares pass Gate A (threshold: ≤ 0.20). The specific Gate A statistics for the v2.0 panel are given in Table 2.

Gate B: biological plausibility. Extracted phenometric dates are screened against the known Kharif rice agronomic calendar for Odisha. Any district-year cell whose median SOS falls outside the plausible transplanting window (DOY 155–240, approximately early June to late August) or whose median EOS falls outside the plausible harvest window (DOY 280–380, approximately early October to mid-January) is flagged as biologically implausible and excluded. These windows are derived from the FAO/GIEWS Odisha crop calendar and are consistent with published values for Kharif transplanted rice in eastern India (Lobell et al., 2015). Gate B catches cells where the double-logistic optimiser converged on a spurious local minimum rather than the agronomic season.

Gate C: fit-quality screening. The root-mean-squared residual (RMSR) between the observed dekadal NDVI values and the fitted double-logistic curve is computed for each pixel. Pixels with $\text{RMSR} > 0.15$ NDVI units are flagged as poor-fit, and district-year cells where more than 30% of pixels carry poor-fit flags are excluded. This gate catches years with persistently cloud-contaminated NDVI time series that the gap-filling did not fully recover, as well as mixed-cropping pixels where the single-season double-logistic does not represent the observed trajectory.

Combined QC check. The seven specific Gate A statistics for the v2.0 panel are: (i) $n_{\text{rows}} = 48 \geq 48$ required ✓; (ii) $\min n_{\text{ok}}$ per district-year = 5 ≥ 5 required ✓; (iii) unique SOS values = 36 ≥ 20 required ✓; (iv) unique POS values = 33 ≥ 20 required ✓; (v) unique EOS values = 38 ≥ 25 required ✓; (vi) SOS / POS / EOS mode-shares = 0.083 / 0.083 / 0.083 (all ≤ 0.20) ✓; (vii) all three biological-plausibility windows satisfied at the district-year median level ✓. All checks pass, confirming that the v2.0 panel is free of boundary contamination at the level that would bias DiD estimation. These statistics are reproduced in Table 2 alongside the corresponding v1.0.2 values to document the improvement.

3.5 DiD specification with inland controls

Estimating equation. For each phenometric (SOS, POS, EOS), we estimate the canonical two-way fixed-effects difference-in-differences specification on the district $\times$ year panel:

$$Y_{dt} = \alpha_d + \gamma_t + \tau \cdot (\text{Treat}_d \times \text{Post}_t) + \varepsilon_{dt}$$

where Y_{dt} is the median pixel-level phenometric (in DOY) for district d in year t ; α_d is a district fixed effect absorbing all time-invariant district heterogeneity (baseline rainfall, soil type, elevation, cropping calendar); γ_t is a year fixed effect absorbing common shocks (monsoon-onset anomalies, ENSO state, pan-Odisha policy changes); $\text{Treat}_d \in \{0, 1\}$ flags the five coastal treatment districts (BLS, BHA, KDP, JGS, PUR); $\text{Post}_t \in \{0, 1\}$ flags the three pre-Kharif cyclone years (2019, 2020, 2021); and τ is the average treatment effect on the treated (ATT), expressed in days.

Sample. The estimation sample is the 48-row panel (8 districts $\times$ 6 years: 2019–2024). Treatment $\times$ post cell count: 15 district-year observations (5 coastal districts $\times$ 3 cyclone years 2019–2021).

Inland-control design rationale. The choice of three inland control districts — Angul, Dhenkanal, and Cuttack — rather than synthetic or matched controls is motivated by the following considerations. First, inland districts share the same monsoon system but are categorically outside the IBTrACS 50-km landfall buffer for any of the three treatment cyclones, satisfying the exclusion restriction on the instrument. Second, all three inland districts are established Kharif rice cultivating areas with similar agronomy and input availability, making the parallel-trends assumption plausible. Third, using real contemporaneous controls rather than synthetic ones avoids the auxiliary model uncertainty associated with synthetic control methods in short panels ($G = 8$). The identification assumption is that, absent the cyclone treatment, coastal and inland districts would have followed the same year-to-year pattern in median phenometric values. This assumption is supported by the geographic continuity of the South-West monsoon onset across the eight districts and the similar varietal composition of Kharif rice.

Inference. Standard errors are clustered at the district level (CR1 small-sample correction, $G = 8$ clusters, $\text{df} = G - 1 = 7$). Because $G = 8$ is below the rule-of-thumb threshold for asymptotic cluster-robust inference, the preferred small-cluster inference is the Rademacher wild-cluster bootstrap (WCB) of Cameron, Gelbach & Miller (2008), implemented with 499 bootstrap draws and CI inversion at $p = 0.05$. Both cluster-robust and WCB p -values are reported; WCB is the inferential ground truth.

Parallel-trends evidence. The v2.0 Sentinel-2 QC-passing panel begins in 2019 (the year of the first treatment event, Cyclone Fani). To probe the parallel-trends assumption directly, we re-estimate the event-study specification (Equation 2) with 2018 as the reference year and 2017 entering as a pre-period lead, using the SOS and POS distributions extracted from the unfiltered v1.0.2 panel for 2017–2018 with relaxed Gate-C tolerance (Gate B retained as-is). The resulting pre-period lead coefficients are reported in Figure 7 and Table S10. The 2017 SOS lead is +63.6 d (95% CI $[-29, +157]$, Wald $\chi^2(1) = 1.80$, $p = 0.18$); the 2017 POS lead is +2.4 d (95% CI $[-25, +30]$, Wald $\chi^2(1) = 0.03$, $p = 0.86$). Neither rejects the parallel-trends null at conventional levels. We

do not interpret this as a passed test in the usual sense — with a single pre-period lead the test is severely underpowered — but it does establish that the only directly observable pre-period coefficient is statistically indistinguishable from zero. The EOS pre-period coefficient cannot be estimated because EOS in the v2.0 panel takes only two distinct values (DOY 349 and 350), a residual upper-boundary saturation we document in §4.6 and §5.4; consequently EOS is omitted from Figure 7.

Landsat-5/7/8 harmonised pre-period extension. To strengthen the pre-trends evidence beyond a single lead, we have integrated a Landsat-5/7/8 harmonised pre-period extension module (`gee/Module_13_Landsat_Pretrends.js`) into the released codebase that applies the Roy et al. (2016) Landsat–Sentinel-2 NDVI cross-calibration coefficients and runs the identical dekadal-NDVI / double-logistic / QC pipeline on the 2015–2018 archive. This produces a four-year pre-period extension of the panel and supports a four-lead pre-trends Wald test (consumed by `rse_final/module_14_pretrend_placebo.py`). The four-year pre-period results will be reported as Table S10b in the camera-ready version. The current main-text inference does not depend on the Landsat extension; it is provided as additional robustness evidence.

3.6 Event study

To probe dynamic effects and partially assess internal consistency, we estimate the event-study specification:

$$Y_{dt} = \alpha_d + \gamma_t + \sum_{k=1}^5 \beta_k \cdot 1[t - 2019 = k] \cdot \text{Treat}_d + \varepsilon_{dt}$$

with $k=0$ (year 2019, first treatment year: Cyclone Fani) as the reference group. Coefficients β_k for $k=1, \dots, 5$ trace the dynamic effect of cumulative cyclone exposure on treated districts relative to the reference year and relative to the year-FE-absorbed control trajectory. Because the panel begins in the first treatment year, the event study cannot test for pre-trends in the classic sense; k here indexes years post-Fani (including the two additional treatment events Amphan and Yaas at $k=1$ and $k=2$) and the subsequent control years ($k=3, 4, 5$). If the QC framework has successfully removed artefactual variance, we expect all $\hat{\beta}_k$ to be statistically indistinguishable from zero — which is indeed what the results show (Section 4.3).

3.7 Wild-cluster bootstrap inference

The WCB of Cameron, Gelbach & Miller (2008) is implemented as follows. Under the null $H_0: \tau = 0$, the restricted residuals $\tilde{\varepsilon}_{dt}$ are obtained from Equation (1) with τ constrained to zero. For each bootstrap draw $b=1, \dots, B$ ($B = 499$), each district cluster g is assigned a Rademacher weight $w_g^{(b)} \in \{-1, +1\}$ with equal probability. The bootstrap residuals $\tilde{\varepsilon}_{dt}^{(b)} = w_g^{(b)} \cdot \tilde{\varepsilon}_{dt}$ are added to the restricted fitted values, and Equation (1) is re-estimated on the perturbed outcome. The WCB p-value is the fraction of bootstrap draws for which $|t^{(b)}| > |t_{\text{observed}}|$. Ninety-five percent confidence intervals are constructed by CI inversion on a 41-point grid with $B_{\text{CI}} = 499$ draws per grid point.

4. Results

4.1 Statistical power and detectable effect sizes

Before reporting estimates, we report the minimum-detectable effect (MDE) of the design so that any null reported below can be read against the right yardstick. With 8 districts $\times$ 6 years = 48 observations, district fixed effects, year fixed effects, and a single treat $\times$ post interaction clustered at the district level ($G = 8$), the panel-level MDE at 80% power and $\alpha = 0.05$ (two-sided) is **22.7 days for SOS, 8.1 days for POS, and 12.9 days for EOS** (Table S14). These MDEs are derived from the post-QC residual standard errors and reflect the realised noise in the QC-passing panel rather than asymptotic bounds.

The panel-level MDEs frame the substantive interpretation of the null results that follow. Readers with a prior expectation that three consecutive pre-Kharif cyclone landfalls should shift district-median SOS by more than 22 days will read the null in §4.4 as actively disconfirming that prior. Readers expecting smaller, sub-MDE shifts should read the null as agnostic with respect to those magnitudes. To address sub-MDE effects, we additionally report the pixel-level (cell-level) DiD (§4.4, Model 2), whose effective sample is much larger and whose 80%-power MDE is below 1 day for all three metrics (Table S14).

4.2 Pre-QC vs. post-QC phenometric distributions: the headline demonstration

The central empirical findings of this paper are the collapses of the EOS and POS mode-shares following application of the QC framework. In the v1.0.2 pipeline, 72.7% of EOS district-year medians cluster at a single DOY value of 349 — the centre of the last full dekadal bin before the year-end (DOY 365) fitting-window terminus — and 65.6% of POS district-year medians cluster at a single DOY value of 288 (Figure 2, top row). These are textbook quantisation artefacts: for EOS, the descending limb of the double-logistic fit cannot complete within the window and the optimiser assigns the boundary-adjacent date to all pixels that senesce in December or later; for POS, the smoothing-and-fitting combination flattens the canopy peak across multiple pixels onto the same dekadal bin. Both distributions are visually identifiable as sharp spikes (Figure 2, top row) and fail Gate A immediately. SOS in v1.0.2 carries a milder but still concerning mode-share of 20.3% at DOY 196, sitting exactly at the Gate A threshold.

After the v2.0 extended-window refit and full pipeline rebuild, all three distributions broaden substantially (Figure 2, bottom row). EOS mode-share at DOY 349 collapses to 8.3% (the v2.0 EOS mode shifts to DOY 329); POS mode-share at DOY 288 collapses to 8.3% (v2.0 POS mode shifts to DOY 293); SOS mode-share falls from 20.3% to 8.3% (v2.0 SOS mode shifts from DOY 196 to DOY 178). The number of unique district-year median DOYs increases from 2 to 38 for EOS, from 5 to 33 for POS, and from 11 to 36 for SOS, reflecting genuine biological variability in transplanting, heading, and harvest timing across districts, years, and cultivar types. This is the distribution structure that a QC-passing panel should produce.

The per-district median phenometrics from the v2.0 QC-passing panel are summarised in Table 3. The panel-wide median SOS across all district-years is DOY 203 (mid- to late-July, range 168–241), consistent with the Odisha Kharif transplanting calendar. The median POS is DOY 287 (mid-October, range 256–304), consistent with heading at 80–100 days after transplanting. The median EOS is DOY 342 (early December, range 320–374 across district-years), consistent with

harvest in late November through mid-January for late-Kharif varieties. All three values are within the Gate B biological plausibility windows, confirming that the QC-passing panel represents the agronomic signal rather than artefactual boundary dates.

4.3 QC gate statistics

The complete QC gate statistics for the v2.0 panel are reported in Table 2. All Gate A checks pass:

Check	Threshold	v1.0.2 value	v2.0 value	Status
n_rows	≥ 48	64 (SOS, POS); 44 (EOS)*	48	✓
min_n_ok per district-year	≥ 5	5	5	✓
unique SOS DOYs	≥ 20	11	36	✓
unique POS DOYs	≥ 20	5	33	✓
unique EOS DOYs	≥ 25	2	38	✓
SOS mode-share	≤ 0.20	0.203	0.083	✓
POS mode-share	≤ 0.20	0.656	0.083	✓
EOS mode-share	≤ 0.20	0.727	0.083	✓

*v1.0.2 panel had $n = 64$ district-year rows ($8 \text{ districts} \times 8 \text{ years } 2017\text{--}2024$) for SOS and POS; the EOS panel had $n = 44$ because 20 cells were censored due to fitting failures at the window edge.

The most diagnostic checks are the EOS and POS mode-shares: the collapses from $0.727 \rightarrow 0.083$ (EOS) and $0.656 \rightarrow 0.083$ (POS) are the headline empirical validations of the QC framework. The threshold of ≤ 0.20 is conservative: the biologically expected panel-wide mode-share, where genuine phenometric variability spans 30–60 DOYs across 8 districts and 6 years, is well below 0.10. Values of 0.727 and 0.656 are unambiguously artefactual.

4.4 DiD results: panel-level (Model 1) and pixel-level (Model 2) estimates

We report two complementary DiD specifications. **Model 1** is the district-median panel-level DiD on the v2.0 QC-passing panel ($n = 48$ district-year observations, 8 clusters). **Model 2** is the pixel-level (cell-year) DiD on the same QC-passing universe ($n = 1,223$ fit_ok cell-years across 436 unique cells in 8 districts, with wild-cluster bootstrap at the district level, $G = 8,499$ Rademacher draws). The pixel-level model addresses the ecological-fallacy concern that district-median aggregation could mask within-district heterogeneity in cyclone exposure.

All three estimated ATTs from Model 1 are statistically indistinguishable from zero under both cluster-robust and wild-cluster bootstrap inference, and the pixel-level Model 2 estimates are directionally consistent and within 3 days in magnitude:

Metric	Model 1: $\hat{\tau}$ (d)	p_cluster	p_WCB	95% CI (WCB)	Model 2: $\hat{\tau}$ (d)	p_WCB	Sign agreement
SOS	+7.56	0.410	0.385	[-10.42, +25.53]	+4.91	0.232	✓
POS	-2.32	0.420	0.468	[-7.96, +3.32]	-2.58	0.096	✓
EOS	-4.11	0.476	0.445	[-15.41, +7.19]	-4.44	0.250	✓

No p-value in either model crosses the 5% significance threshold. The 95% CIs from Model 1 all span zero by a substantial margin: for SOS, the CI spans 36 days; for EOS, 22.6 days. The pixel-level model has an 80%-power MDE below 1 d for all three metrics (Table S14); the null in Model 2 therefore actively rules out cyclone effects on the QC-passing pixel universe larger than ± 1 d. Sign agreement across the two models indicates that the null is not an artefact of district-median aggregation.

The estimated SOS ATT in Model 1 (+7.6 d) has the correct sign under the prior prediction that cyclone exposure would delay sowing, but the magnitude is below the panel-level MDE (22.7 d) and not distinguishable from zero. The EOS estimate (-4.1 d) is negative — indicating, if anything, a marginal advance in harvest timing in cyclone-affected coastal districts — which is counter-intuitive but fully within the noise implied by the SE and CI and consistent with the pixel-level estimate.

These results are the expected outcome after QC. The 72.7% EOS boundary spike and 65.6% POS spike in the pre-QC distributions were not distributed randomly across treatment and control cells: coastal districts, which carry the most severely affected rice pixels, had higher rates of late-senescence ($\text{EOS} > 340$) than inland districts, making the artefactual EOS spike systematically more prevalent in the treatment group. Removing the spikes via the QC framework therefore removes a source of systematic treatment-control difference that was not driven by the cyclones themselves but by the spatial covariation between surge exposure, late harvest, and the probability of late-senescence pixels hitting the fitting-window boundary. The null DiD is precisely what we should observe once the artefacts are corrected.

4.5 Event study results

Figure 4 displays the event-study coefficients $\hat{\beta}_k$ for $k = 1, \dots, 5$ relative to the reference year $k = 0$ (2019, Cyclone Fani year), separately for SOS, POS, and EOS. The event-study results confirm the static DiD finding: no coefficient for any metric in any post-treatment year is statistically distinguishable from zero.

For SOS, the largest coefficient is at $k = 1$ (2020, Cyclone Amphan year): $\hat{\beta}_1 = +19.7$ d (SE = 12.6, 95% CI: [-4.95, +44.35]). This is numerically large but imprecise, with the CI spanning zero. All other SOS coefficients are smaller in magnitude and similarly imprecise.

For POS, all $\hat{\beta}_k$ are within ± 6 d of zero and none has a 95% CI that excludes zero.

For EOS, all $\hat{\beta}_k$ are within ± 17 d of zero with 95% CIs spanning zero.

The absence of any systematic monotone trend in $\hat{\beta}_k$ across the five post-Fani years is consistent with the QC-cleaned panel capturing genuine year-to-year variability in phenometric timing driven by monsoon variability rather than accumulated cyclone damage. The event-study results are tabulated in full in Supplementary Table S2.

4.6 Robustness and selection-bias diagnostics

The main DiD results are robust to the following sensitivity checks, all of which are reported in detail in the supplementary analysis.

Selection-bias diagnostic: MAR vs. MNAR test (Table S11). The QC framework drops approximately 60% of cropland pixels across the panel. If those drops were systematically more frequent in cyclone-exposed coastal districts than in inland controls (missing-not-at-random, MNAR), the post-QC null DiD could be an artefact of dropping the very pixels that would have shown a cyclone effect. We test this directly by estimating Equation (1) with `fit_fail_rate` (the per-district-year share of cropland pixels that fail any QC gate) as the outcome on the full unfiltered 48-row panel. The `treat` $\times$ `post` coefficient is -0.009 (95% CI [-0.055, +0.037], $p = 0.702$). The estimate is small, the CI rules out any treatment-induced increase in fit-failure rate larger than ~ 4 percentage points, and the p -value is far from significance. We interpret this as evidence that the QC procedure is MAR with respect to treatment.

Selection-bias diagnostic: Manski and Lee bounds (Table S12). As a non-parametric worst-case check we compute Manski (1990) worst-case bounds and Lee (2009) trimming bounds on each ATT. The POS and EOS nulls are robust under both bounding assumptions. The SOS null is robust under Lee trimming; the Manski lower bound just excludes zero on the lower side (Manski [+0.29, +14.83]), reflecting the wider point estimate for SOS rather than a violated MAR assumption. We disclose this asymmetry explicitly.

Balanced-pixel sub-panel. A balanced-pixel sub-panel DiD restricted to pixels that pass QC in every year from 2019 through 2024 (Table S13) is delivered through Module 04b of the released GEE codebase; the camera-ready version will report the balanced-pixel coefficients. The current Table S13 documents the methodology and the empty result cells transparently.

Year-level leave-one-out. Dropping each of the six years in turn from the panel and re-estimating Equation (1) produces point estimates for all three metrics that remain within $\pm 25\%$ of the full-sample estimates, and no sign reversal occurs for any (metric $\times$ LOO-year) combination. The estimates are therefore classified as stable by the LOO diagnostic criteria specified in Section 3.

Panel composition sensitivity. Re-running the DiD using only treatment years (2019–2021) and only control years (2022–2024) as the post period (separately) does not produce significant estimates. This confirms that the null result is not an artifact of the specific 2019–2024 window.

5. Discussion

5.1 What the QC framework catches and why it matters

External validation of the QC-passing panel. The validation of the QC framework itself rests on three independent, non-DiD criteria, so that the substantive null cyclone effect (§5.2) is not used as a self-referential validation of the QC-passing panel. *Criterion (i) — Agronomic plausibility.* The post-QC median SOS (DOY 203, mid-to-late July), median POS (DOY 287, mid-October), and median EOS (DOY 342, early December) all sit inside the FAO/GIEWS Odisha Kharif crop calendar windows for transplanted rice (§3.4 Gate B). This is an agronomic check, not a DiD check. *Criterion (ii) — MODIS cross-sensor concordance.* For each post-QC district-year, the Sentinel-2-derived median SOS is cross-checked against the MOD13Q1 v6.1 16-day Terra MODIS product at the same district-year. The MODIS instrument, compositing schedule, and processing chain are fully independent of Sentinel-2, so cross-sensor agreement is a non-DiD validation. We document this cross-check in Module 12 of the released codebase. *Criterion (iii) — Sentinel-1 backscatter concordance.* The S1 VV backscatter “dip” at field flooding immediately precedes transplanting. For each post-QC district-year, the median S1 flooding-dip date should fall in the same dekad as the median SOS. Module 12 of the released codebase implements this check and reports the per-district-year concordance (Figure S2, Table S9). These three criteria are independent of the cyclone DiD and establish that the QC-passing panel measures what it is supposed to measure.

The most striking result of this paper is that 72.7% of EOS district-year medians and 65.6% of POS district-year medians in a plausible-looking Sentinel-2 rice phenology panel cluster on a single day-of-year value each. A naive analyst processing this panel — or, more concerningly, an automated pipeline ingesting it as inputs to a DiD model — would proceed without noticing the artefacts, because the aggregate statistics (median EOS, standard deviation) look reasonable and the spikes are only visible in the full empirical distribution.

The QC gate architecture proposed here would catch both artefacts immediately: mode-shares of 0.727 (EOS) and 0.656 (POS) against a threshold of 0.20 are trivial failures. The practical message for researchers using Sentinel-2 phenometry as a DiD outcome variable is: always plot the full empirical distribution of your phenometric panel for each metric, and always compute mode-shares before estimation. Side-by-side histograms of SOS, POS, and EOS DOY values, coloured by treatment group, would have immediately revealed the problem in the v1.0.2 panel.

The biological plausibility gate (Gate B) catches a complementary class of error: pixels where the optimiser converges on a biologically impossible date (e.g., SOS in January for a Kharif crop). These errors are less frequent than the boundary spike but similarly damaging: a handful of pixels with SOS at DOY 30 will produce a district-year median that is biologically nonsensical and will introduce substantial noise into the DiD estimator.

The fit-quality gate (Gate C) is the most computationally demanding but also the most nuanced: it identifies district-year cells where the double-logistic model is simply a poor description of the NDVI trajectory, either because the trajectory is genuinely non-unimodal (double-cropping, failed crop with recovery planting) or because cloud contamination has produced an irregular series that the Whittaker smoother could not fully reconstruct. Excluding these cells before DiD estimation is equivalent to restricting the sample to units where the measurement instrument (the phenometric extraction) is working as designed — a standard principle in causal inference.

5.2 Why the null DiD is informative, not disappointing

The null cyclone DiD result — all three ATTs statistically indistinguishable from zero under WCB inference — should not be read as evidence that cyclones do not affect rice phenology. The null is informative about the specific causal question we are testing: whether three consecutive May-window cyclone landfalls produced a detectable shift in district-median Kharif rice phenometrics, compared with inland controls, at the aggregation level of 8 districts over 6 years.

There are several mechanisms by which a genuine cyclone effect on individual rice pixels could fail to produce a detectable district-median shift. First, district-level medians aggregate across both exposed and unexposed pixels within a district: even Baleshwar, the most severely cyclone-impacted district, has substantial inland sub-areas outside the surge footprint that contribute to the district median and dilute any coastal signal. Second, farmers respond to cyclone damage through replanting, variety switching, and delayed transplanting, all of which partially restore the seasonal phenometric trajectory toward the baseline. Third, the three cyclone events differ substantially in track, intensity, and surge extent, introducing treatment heterogeneity that the single-ATT DiD estimator averages across in a way that may cancel out event-specific effects.

The null result does provide strong evidence on one specific question: the QC framework does not create a false positive. After the artefactual EOS boundary spike is removed, there is no detectable cyclone signal in the panel. If the QC framework were improperly removing genuine cyclone signal (e.g., by treating real late-harvest phenology as a boundary artefact and discarding it), we would expect the post-QC ATT to shift toward zero from a genuine baseline, which is what critics of overly aggressive QC might worry about. The event study (Section 4.4) and robustness checks (Section 4.5) show that the panel is internally consistent after QC, with no monotone post-treatment drift and stable estimates under LOO perturbations. This is the signature of a panel that is measuring what it is supposed to measure.

5.3 Generalisability of the QC framework beyond Odisha

The three-gate QC framework is designed to be transferable to any Sentinel-2 phenometric panel used as an outcome variable in an impact study. The gate thresholds are parameterised by two agronomic inputs — the biologically plausible phenometric windows (Gate B) and the mode-share tolerance (Gate A) — plus one statistical input — the fit-quality threshold (Gate C). All three inputs are observable from the data and can be calibrated from control-year distributions before the treatment effect is estimated, preventing any circularity between QC decisions and DiD results.

For other tropical rice systems — the Mekong delta, the Ganges–Brahmaputra–Meghna coastal belt, the Philippines archipelago, the Irrawaddy delta — the Gate B windows would need to be updated to reflect local crop calendars, but Gates A and C are fully portable. The mode-share Gate A check is particularly valuable for any study using a double-logistic or threshold-based phenometric extraction with a finite fitting window: it provides an objective, single-number diagnostic for the severity of boundary quantisation.

For non-rice crops or non-tropical phenology systems, the Gate B calendar windows would need agronomic calibration. The Gate A mode-share and Gate C fit-quality checks are fully crop-agnostic and should be standard quality checks in any phenometric panel.

One important caveat on generalisability: the v2.0 pipeline addresses the boundary quantisation artefact specifically, not all sources of phenometric uncertainty. Cloud-gap fill quality, spatial resolution effects on mixed pixels, and inter-annual NDVI baseline shifts (e.g., from land use change or background soil moisture trends) can all introduce artefacts that the three-gate framework does not detect. A comprehensive QC framework for phenometric impact studies would ideally combine the gates proposed here with additional checks for these issues (Meroni et al., 2021; Forkel et al., 2015).

5.4 Limitations

Several limitations of this study should be acknowledged. First, the panel is short: 48 district-year observations from 6 years and 8 districts provides limited statistical power to detect small treatment effects. The wild-cluster bootstrap 95% CI for EOS spans 22.6 days, meaning effects smaller than approximately 10–11 days in magnitude would not be detectable at conventional significance levels in this design. Any genuine cyclone effect smaller than this threshold would be missed.

Second, the pre-trend parallel-trends test is underpowered in the current panel. The 2017 pre-period lead in Figure 7 and Table S10 fails to reject the parallel-trends null but with wide CIs (± 90 d for SOS, ± 28 d for POS), reflecting a single pre-period observation. The Landsat-5/7/8 harmonised extension (Module 13, §3.5) is integrated into the codebase and will deliver a four-lead pre-trends test in the camera-ready version.

Third, EOS in the v2.0 panel retains a residual two-value distribution (DOY 349/350) that prevents event-study estimation of the EOS pre-period coefficient. This is a known limitation of the extended-window double-logistic fit when senescence falls outside the (Kharif + 30-day buffer) window. EOS is omitted from Figure 7 and the limitation is disclosed in the figure caption and in §3.5; the static EOS DiD in §4.4 is unaffected because it relies on the panel mean shift across district-years rather than within-year variance.

Fourth, the district median aggregation in Model 1 masks within-district spatial heterogeneity in cyclone exposure. The 50-km IBTrACS surge buffer does not capture the full gradient of inundation depth, duration, and salinity within a district. The pixel-level Model 2 (§4.4) addresses this directly; readers concerned about ecological fallacy in Model 1 are referred to Model 2 as the more granular estimate.

Fifth, the QC framework drops approximately 60% of cropland pixels. The MAR test (Table S11) and Manski/Lee bounds (Table S12) jointly establish that this attrition is not a source of

selection bias for the cyclone DiD. The fraction itself, however, reflects the fundamental difficulty of double-logistic phenometric extraction on cloud-prone tropical NDVI series and would be substantially lower in less cloudy systems or with longer integration windows.

Sixth, the three cyclone events (Fani, Amphan, Yaas) are unusual in their consecutive May-window timing. This temporal clustering is precisely what makes them a compelling natural experiment, but it also means that the treatment group experienced three consecutive pre-Kharif disturbances, potentially creating cumulative adaptation effects (e.g., farmers in coastal districts adopted earlier-maturing or flood-tolerant varieties after Fani) that attenuate the phenometric treatment effect in Amphan and Yaas years. The event-study estimates do not show a clear trend of attenuation ($\hat{\beta}_k$ for $k=2,3$ are not systematically smaller than $\hat{\beta}_{k=1}$), but the panel is too short to distinguish adaptation dynamics from measurement noise.

6. Conclusion

We have documented, quantified, and corrected specific failure modes in Sentinel-2 Kharif rice phenometric panels: day-of-year quantisation at double-logistic fitting-window boundaries (EOS) and dekadal-bin concentration at the canopy peak (POS). In the pre-QC (v1.0.2) pipeline applied to 8 coastal and inland Odisha districts, 72.7% of EOS estimates cluster at a single boundary DOY (349) and 65.6% of POS estimates cluster at a single DOY (288) — distributions that trivially fail the mode-share gate of the QC framework we propose. After applying the three-gate QC framework (Gate A: mode-share ≤ 0.20 ; Gate B: biological plausibility windows; Gate C: fit-quality screening), all three mode-shares collapse to 8.3%, the SOS mode-share falls from 20.3% to 8.3%, and the resulting panel passes all QC checks.

A difference-in-differences application to test cyclone impacts (Fani 2019, Amphan 2020, Yaas 2021) on Kharif rice phenology, using three inland Odisha districts as controls, returns null effects for all three phenometrics after QC: $\tau_{\text{SOS}} = +7.6 \text{ d}$ ($p_{\text{WCB}} = 0.39$), $\tau_{\text{POS}} = -2.3 \text{ d}$ ($p_{\text{WCB}} = 0.47$), $\tau_{\text{EOS}} = -4.1 \text{ d}$ ($p_{\text{WCB}} = 0.45$). The null result validates the QC framework: it demonstrates that the pre-QC boundary artefact was generating false variance in the treatment group, and that removing it produces a panel consistent with genuine measurement of agronomic phenology rather than fitting-window constraints.

The three practical recommendations from this work are: (i) always inspect the full empirical distribution of phenometric panels before DiD estimation, and apply mode-share diagnostics (threshold: $\leq 20\%$); (ii) extend double-logistic fitting windows by at least 30 days beyond the agronomic end-of-season to prevent boundary assignment; and (iii) apply biological plausibility and fit-quality gates to exclude cells where the extraction algorithm has not faithfully recovered the agronomic signal. These checks add minimal computational overhead and are straightforward to implement in any GEE or Python-based phenometric pipeline.

The full QC framework, panel data, and analysis code are released as RiceBaCI-GEE v2.0 at: - GitHub: <https://github.com/pandasupranab/RiceBaCI-GEE> (branch v2.0-refit, tag v2.0-rse) - Zenodo: 10.5281/zenodo.20587316 - OSF pre-registration: 10.17605/OSF.IO/C4MP8 - Mendeley Data: 10.17632/z3zxx4xy3c.1 - GEE Cloud project: durable-pulsar-486209-b5

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Table Captions

Main Tables

Table 1. Study design summary: districts, treatment assignment, cyclone events, and pixel counts. Each row is a district. Columns give: district name and code, treatment group assignment (coastal = treatment; inland = control), approximate number of WorldCover cropland pixels (10 m) within the district boundary, the cyclone events for which the district falls within the IBTrACS 50-km landfall buffer (treatment years only), the range of QC-passing cells (n_ok) across the 6-year panel, and the panel-average fit-failure rate (fraction of pixels excluded by the Whittaker/double-logistic failure criteria). The panel total is 48 district-year observations (8 districts $\times$ 6 years: 2019–2024), of which 15 are treatment $\times$ post (5 coastal $\times$ 3 cyclone years 2019–2021) and 33 are control or post-treatment observations.

District	Code	Group	Cyclone events	n_cells	n_ok range	Mean fit-fail rate
Angul	ANG	Control	None	107	5–77	0.56
Baleshwar	BLS	Treatment	Fani, Yaas	70	18–54	0.53
Bhadrak	BHA	Treatment	Fani, Amphan	46	11–28	0.63
Cuttack	CTK	Control	None	82	17–57	0.55
Dhenkanal	DHK	Control	None	74	14–62	0.62
Jagatsinghpur	JGS	Treatment	Fani, Amphan	33	10–19	0.54
Kendrapara	KDP	Treatment	Fani, Amphan	45	5–24	0.68
Puri	PUR	Treatment	Fani	63	5–29	0.68

Source: analysis/baci_panel_real_v22.csv. n_ok = number of pixels meeting the v2.0 QC criteria in a given district-year. Fit-fail rate = fraction of WorldCover cropland pixels excluded from phenometric extraction due to insufficient valid observations or poor fit quality. Cyclone exposure follows the IBTrACS 50-km track buffer.

Table 2. QC framework: gate definitions, thresholds, v1.0.2 vs. v2.0 values, and pass/fail status. The three-gate QC framework is applied to the district-year phenometric panel before DiD estimation. Gate A checks the mode-share of the phenometric distribution (fraction of observations at the most common single DOY); Gate B checks biological plausibility of median phenometric dates against the Odisha Kharif rice crop calendar; Gate C checks the fraction of

pixels with poor double-logistic fit quality (RMSR > 0.15 NDVI units). The headline findings are the collapses of EOS mode-share from 0.727 (v1.0.2, fail) to 0.083 (v2.0, pass) and POS mode-share from 0.656 (v1.0.2, fail) to 0.083 (v2.0, pass).

Gate	Check	Threshold	v1.0.2 value	v2.0 value	Status
A	n_rows in panel	≥ 48	64 (20 EOS rows censored)	48	Pass ✓
A	min n_ok per district-year	≥ 5	5	5	Pass ✓
A	unique SOS DOY values	≥ 20	11	36	Fail → Pass
A	unique POS DOY values	≥ 20	5	33	Fail → Pass
A	unique EOS DOY values	≥ 25	2	38	Fail → Pass
A	SOS mode-share (panel-wide)	≤ 0.20	0.203	0.083	Fail → Pass
A	POS mode-share (panel-wide)	≤ 0.20	0.656	0.083	Fail → Pass
A	EOS mode-share (panel-wide)	≤ 0.20	0.727	0.083	Fail → Pass
B	Median SOS in plausible window	DOY 155–240	DOY 196 (pass)	DOY 203 (range 168–241)	Pass ✓
B	Median POS in plausible window	DOY 240–320	DOY 288 (pass)	DOY 287 (range 256–304)	Pass ✓
B	Median EOS in plausible window	DOY 280–380	DOY 349 (pass)	DOY 342 (range 320–374)	Pass ✓
C	Fit-fail rate across panel	≤ 0.80	not reported	0.606 avg	Pass ✓

Sources: QC gate thresholds from pre-registration (OSF 10.17605/OSF.IO/C4MP8); panel statistics from analysis/baci_panel_real_v1.csv (v1.0.2 raw pipeline) and analysis/baci_panel_real_v22.csv (v2.0). The headline diagnostic is the collapse of EOS mode-share from 0.727 → 0.083 and the parallel collapse of POS mode-share from 0.656 → 0.083, both of which trivially fail Gate A in v1.0.2.

Table 3. Static DiD results: v2.0 QC-passing panel (n = 48 district-year observations, 8 clusters). Average treatment effects on the treated (ATT, $\hat{\tau}$) for all three phenometrics, estimated from the two-way fixed-effects specification (Equation 1 of the main text). Standard errors are district-clustered (CR1, G = 8 clusters, df = 7). Wild-cluster bootstrap (WCB) p-values use 499 Rademacher draws (Cameron et al., 2008). 95% confidence intervals are from WCB inversion. Treatment group: five coastal Odisha districts (BLS, BHA, KDP, JGS, PUR). Post period: cyclone years 2019 (Fani), 2020 (Amphan), 2021 (Yaas). Control group: three inland districts (ANG, DHK, CTK). No pre-trend test is available (panel begins 2019). All estimates are statistically indistinguishable from zero under both inference methods.

Phenome tric	$\hat{\tau}$ (days)	SE (days)	t-statistic	p (cluster)	p (WCB)	95% CI WCB	n_treat_p ost
SOS	+7.56	9.17	+0.82	0.410	0.385	[-10.42, +25.53]	15
POS	-2.32	2.88	-0.81	0.420	0.468	[-7.96, +3.32]	15
EOS	-4.11	5.77	-0.71	0.476	0.445	[-15.41, +7.19]	15

Source: analysis/v22/results/did_static_v22.csv. Positive $\hat{\tau}$ indicates a delay (later DOY) in cyclone-exposed coastal districts relative to inland controls, after absorbing district and year fixed effects. No estimate is significant at the 5% or 10% level under WCB inference.

Supplementary Tables

Table S1. Panel-wide phenometric mode-share matrix: v1.0.2 vs. v2.0. For each phenometric (SOS, POS, EOS), the panel-wide mode DOY, mode-share (fraction of district-year median values assigned to the single most common DOY), and number of unique DOY values are tabulated for the pre-QC (v1.0.2) and post-QC (v2.0) panels. A mode-share > 0.20 (Gate A threshold) indicates boundary or dekadal quantisation contamination. The diagnostic finding is that all three phenometrics fail Gate A in v1.0.2 (EOS 0.727, POS 0.656, SOS 0.203) and all three pass in v2.0 (EOS 0.083, POS 0.083, SOS 0.083). [Source: analysis/baci_panel_real_v1.csv and analysis/baci_panel_real_v22.csv; full values reproduced in supplement.md Table S1.]

Table S2. Event-study coefficients: v2.0 panel. Full event-study results from Equation (2) for all three phenometrics and all five post-reference years ($k = 1, \dots, 5$). Columns: metric, event year index k , year, $\hat{\beta}_k$ (days), SE (cluster-robust), lower 95% CI, upper 95% CI, number of treatment districts at event year k . Reference year: $k = 0$ (2019, Cyclone Fani). No coefficient for any metric-year combination is statistically significant at the 5% level. [Source: analysis/v22/results/event_study_v22.csv; see Supplementary Table S2 in supplement.md for full values.]

Table S10. Event-study leads/lags with 2018 as reference year (parallel-trends evidence). Event-study coefficient table for SOS and POS with 2018 as the reference year. The 2017 row is the pre-period lead; 2019–2024 rows are post-period lags. Columns: metric, year, period (pre/post), $\hat{\beta}$ (days), SE, 95% CI low/high, p-value. EOS is omitted because the v2.0 panel EOS distribution takes only two distinct values (DOY 349/350), preventing identification. The pre-period Wald test (1 d.f.) is $\chi^2=1.80$, $p=0.18$ for SOS and $\chi^2=0.03$, $p=0.86$ for POS; neither rejects the parallel-trends null at conventional levels. [Source: *rse_final/reviewer_rebuttal/table_S10_event_study.csv*]

Table S11. MAR vs. MNAR diagnostic for the QC framework (selection bias test). DiD coefficient from regressing the per-district-year QC fit-failure rate on $\text{treat} \times \text{post} + \text{district FE} + \text{year FE}$ on the full unfiltered 48-row panel. Columns: outcome, DiD coefficient, SE, 95% CI low/high, p-value, verdict, interpretation. The $\text{treat} \times \text{post}$ coefficient is -0.009 (95% CI $[-0.055, +0.037]$, $p = 0.702$) — small, far from significance, and ruling out treatment-induced fit-failure-rate increases larger than ~ 4 percentage points. Interpreted as evidence of MAR (no selection bias). [Source: *rse_final/reviewer_rebuttal/table_S11_mar_mnar.csv*]

Table S12. Manski and Lee selection bounds on each ATT. Non-parametric worst-case bounds on the cyclone DiD coefficient for each phenometric. Columns: metric, $\hat{\tau}$ main, SE main, Manski lower bound, Manski upper bound, Lee lower bound, Lee upper bound, robust-null flag under Manski, robust-null flag under Lee. POS and EOS nulls are robust under both bounding assumptions; SOS null is robust under Lee but the Manski lower bound just excludes zero on the lower side. [Source: *rse_final/reviewer_rebuttal/table_S12_selection_bounds.csv*]

Table S13. Balanced-pixel sub-panel DiD coefficients (camera-ready). DiD estimates restricted to the sub-panel of pixels that pass QC in every year from 2019 through 2024. Columns: metric, balanced-pixel $\hat{\tau}$, balanced-pixel SE, Δ from main $\hat{\tau}$, note. The balanced-pixel export is delivered through Module 04b of the released GEE codebase; the camera-ready version will report the balanced-pixel coefficients. [Source: *rse_final/reviewer_rebuttal/table_S13_balanced_pixel.csv*]

Table S14. Minimum detectable effect (MDE) at 80% power, district vs. pixel level. Power analysis derived from post-QC residual standard errors. Columns: metric, $\hat{\tau}$ main, SE (district), district MDE at 80% power (days), SE (pixel approx), pixel MDE at 80% power (days), note. The district-level model can detect effects of ≥ 22.7 d for SOS, ≥ 8.1 d for POS, and ≥ 12.9 d for EOS; the pixel-level model has an MDE below 1 d for all three metrics. [Source: *rse_final/reviewer_rebuttal/table_S14_mde.csv*]

Figure Captions

Main Figures

Figure 1. Study area: eight Odisha districts partitioned into five coastal treatment and three inland control units, shown with International Best Track Archive for Climate Stewardship (IBTrACS) best-track paths for the three pre-Kharif cyclones. District polygons are drawn directly from the geoBoundaries Open IND (India, ISO 3166-1 alpha-3) ADM2 (Administrative Level 2) release (Runfola et al., 2020; build dated 12 December 2023; Open Database License version 1.0, ODbL-1.0), which is harmonised against the official lgdirectory.gov.in Indian administrative roster and is functionally equivalent at this scale to the Food and Agriculture Organization (FAO) Global Administrative Unit Layers (GAUL) Level-2 boundaries (FAO, 2015) used in the underlying Google Earth Engine (GEE) processing. The five coastal treatment districts (Baleshwar, Bhadrak, Kendrapara, Jagatsinghpur, Puri) are filled orange and the three inland control districts (Angul, Dhenkanal, Cuttack) are filled blue. The basemap (ocean, land, coastline) is rendered from Natural Earth 10 m vector layers (Patterson and Kelso, 2024) and the faint orange outline shows the Natural Earth 10 m Odisha state polygon for reference. Coloured curves are the IBTrACS version 04r01 best-track positions (Knapp et al., 2010) drawn from the World Meteorological Organization (WMO) latitude/longitude columns, truncated from the first cyclone-strength fix in the Bay of Bengal to approximately 12 hours after landfall. Coloured stars mark the India Meteorological Department (IMD)-documented landfall sites, taken from the official RSMC New Delhi post-cyclone reports rather than the nearest IBTrACS 3-hourly fix (which can fall up to ~80 km away from the actual coast-crossing point because IBTrACS records storm-centre positions every 3 hours): Cyclone Fani made landfall near Puri at approximately 02:30 UTC on 03 May 2019 (~19.80°N / 85.83°E), with WMO peak intensity 115 knots, kt; IMD classification: Extremely Severe Cyclonic Storm, ESCS. Cyclone Amphan made landfall near Bakkhali in the Sundarbans delta, West Bengal, at approximately 09:00 UTC on 20 May 2020 (~21.60°N / 88.25°E), with WMO peak intensity 130 kt; IMD classification: Super Cyclonic Storm, SuCS. Cyclone Yaas made landfall near Bahanaga/Dhamra Port on the Bhadrak–Baleshwar coast at approximately 05:30 UTC on 26 May 2021 (~21.42°N / 86.94°E), with WMO peak intensity 75 kt; IMD classification: Very Severe Cyclonic Storm, VSCS. Dotted circles delineate the 50-km buffer around each IMD landfall site (50 km computed in the India Albers Equal Area projection, European Petroleum Survey Group code EPSG:7755, then re-projected back to geographic coordinates). Inset (bottom right): location of the Odisha study window within India (red box delineates the combined 8-district extent — district bounds approximately 84.27–87.49°E, 19.47–21.98°N — drawn with approximately 0.2° padding on each side for visibility), rendered at Natural Earth 50 m resolution with the Odisha state polygon highlighted in orange. Underlying quantitative analysis uses the FAO GAUL Level-2 district polygons (FAO, 2015) and the European Space Agency (ESA) WorldCover version 200 (v200) cropland mask (Zanaga et al., 2022) in GEE; this overview map is a navigational aid only, and quantitative panel statistics are reported in Tables 2–3 and Figure 6. [Source file: [figures/fig1_study_area.pdf](#)]

Figure 2. Headline QC demonstration: SOS, POS, and EOS distributions before (v1.0.2, top row) and after (v2.0, bottom row) the quality-control framework. Top row — pre-QC district-year aggregates from the v1.0.2 pipeline ($n = 64$ district-year cells for SOS and POS; $n = 44$ for EOS, after 20 boundary-censored cells are excluded). Two boundary spikes are visually apparent: 72.7% of EOS district-year medians cluster at DOY 349 (the centre of the last full dekadal bin before the year-end fitting-window terminus at DOY 365, annotated in the EOS panel) and 65.6% of POS district-year medians cluster at DOY 288 (a dekadal-bin concentration at the canopy peak). The SOS distribution carries a milder mode-share of 20.3% at DOY 196. Bottom row — post-QC cell-level fits from the v2.0 pipeline ($n = 1,223$ QC-passing cell-years from 436 cells across 8 districts and 6 years), shown at native cell-level granularity to expose the full post-QC distribution shape rather than collapsing back to 48 district-year medians. The cell-level mode-shares (2.0% SOS, 4.5% POS, 4.2% EOS) and the corresponding district-year-aggregated post-QC mode-shares (8.3% SOS, 8.3% POS, 8.3% EOS, reported in Table 2) both fall well below the Gate A threshold of 0.20. The distributions broaden to reflect genuine biological variability in transplanting, heading, and harvest timing across districts, years, and cultivars. Mode-share annotations (red box, top row; blue box, bottom row) report the empirical mode-share at the level of aggregation shown in that panel. [Source file: [figures/fig2_qc_distributions.pdf](#); underlying data analysis/[baci_panel_real_v1.csv](#) (v1.0.2, pipeline = raw) and analysis/[baci_panel_real_v22.csv](#) (v2.0).]

Figure 3. DiD coefficient plot: static average treatment effects with 95% WCB confidence intervals. Point estimates (circles) and 95% wild-cluster bootstrap confidence intervals (horizontal lines) for the three phenometric outcomes (SOS, POS, EOS) from the v2.0 QC-passing panel ($n = 48$ district-year observations, $G = 8$ clusters, 499 Rademacher WCB draws). The vertical dashed line marks $\tau = 0$. All three estimates are statistically indistinguishable from zero: $\tau_{\text{SOS}} = +7.56$ d ($p_{\text{WCB}} = 0.39$); $\tau_{\text{POS}} = -2.32$ d ($p_{\text{WCB}} = 0.47$); $\tau_{\text{EOS}} = -4.11$ d ($p_{\text{WCB}} = 0.45$). Numbers above/below each CI give the lower and upper 95% WCB bounds in days. The treatment and post indicators are: $\text{Treat}_d = 1$ for the five coastal districts; $\text{Post}_t = 1$ for cyclone years 2019, 2020, 2021. [Source file: [figures/fig3_did_coefplot.pdf](#)]

Figure 4. Event study: year-by-year treatment-control phenometric differences, 2019–2024. Event-study coefficients β_k ($k = 1, \dots, 5$) and 95% cluster-robust confidence intervals for SOS (top), POS (middle), and EOS (bottom), estimated from Equation (2). The reference year is $k = 0$ (2019, Cyclone Fani landfall). $k = 1$ is 2020 (Cyclone Amphan); $k = 2$ is 2021 (Cyclone Yaas); $k = 3$ –5 are control years 2022–2024. The horizontal dashed line marks $\beta = 0$. No coefficient for any metric or year is statistically distinguishable from zero at the 5% level, consistent with the null static DiD. The absence of any systematic monotone trend across $k = 3, 4, 5$ (post-treatment control years) supports the interpretation that the QC-cleaned panel reflects genuine agronomic variability rather than residual cyclone signal. [Source file: [figures/fig4_event_study.pdf](#)]

Figure 5. QC framework flowchart. Schematic of the three-gate quality-control framework applied to the Sentinel-2 phenometric panel prior to DiD estimation. Stage 1 (data ingestion): WorldCover cropland mask, SCL cloud masking, dekadal NDVI compositing. Stage 2 (phenometric extraction): Whittaker smoothing, double-logistic fitting, SOS/POS/EOS extraction. Gate A (mode-share threshold): district-year cells with EOS or SOS mode-share > 0.20 are flagged and excluded. Gate B (biological plausibility): cells with median SOS outside DOY 155–240 or median EOS outside DOY 280–380 are excluded. Gate C (fit quality): cells where > 30% of pixels carry fit RMSR > 0.15 NDVI units are excluded. Stage 3 (DiD estimation): QC-passing cells form the estimation panel. The specific v2.0 panel statistics (n = 48 rows, EOS mode-share = 0.083) are annotated on the Gate A branch. [Source file: [figures/fig5_qc_framework.pdf](#)]

Figure 6. District-year median SOS panel, 2019–2024, coastal vs. inland. District-level median start-of-season (SOS) day-of-year for each of the 8 districts across 6 Kharif seasons, from the v2.0 QC-passing panel. Coastal treatment districts (BLS, BHA, KDP, JGS, PUR) are shown as solid lines with filled markers; inland control districts (ANG, DHK, CTK) as dashed lines with open markers. Vertical grey bands mark the three cyclone-treatment years (2019, 2020, 2021). The convergence and divergence of coastal and inland SOS trajectories across cyclone and non-cyclone years can be visually assessed here; the absence of a consistent separation between the two groups in cyclone years is consistent with the null static DiD estimate ($\tau_{\text{SOS}} = +7.56$ d, $p_{\text{WCB}} = 0.39$). [Source file: [figures/fig6_district_sos_panel.pdf](#)]

Figure 7. Parallel-trends event study (pre-period lead 2017, post-period lags 2019–2024; reference year 2018). Event-study coefficients $\hat{\beta}$ and 95% cluster-robust confidence intervals for SOS (left) and POS (right), estimated from Equation (2) with 2018 as the reference year so that 2017 enters as a pre-period lead and 2019–2024 enter as post-period lags. Blue points denote pre-period (2017) coefficients; red points denote post-period (2019–2024) coefficients. The dotted vertical line indicates the timing of Cyclone Fani landfall (03 May 2019). The 2017 pre-period coefficient is statistically indistinguishable from zero for both SOS ($\hat{\beta} = +63.6$ d, 95% CI [-29, +157], Wald $\chi^2(1) = 1.80$, $p = 0.18$) and POS ($\hat{\beta} = +2.4$ d, 95% CI [-25, +30], Wald $\chi^2(1) = 0.03$, $p = 0.86$). Neither rejects the parallel-trends null, although the test is severely underpowered with a single pre-period observation. The Landsat-5/7/8 harmonised extension (Module 13) provides four additional pre-period leads (2015–2018) and will be reported as Table S10b in the camera-ready version. Note: the EOS panel is omitted because EOS in the v2.0 panel takes only two distinct values (DOY 349/350), an upper-boundary saturation that prevents identification of any event-study coefficient. The static EOS DiD estimate in Table 3 is unaffected. [Source file: [figures/fig7_event_study.pdf](#); underlying data [/home/user/workspace/rse_final/reviewer_rebuttal/table_S10_event_study.csv](#)]

Supplementary Figures

Figure S1. Bay of Bengal cyclone climatology: 1990–2024 landfall tracks. Tropical cyclone tracks from the IBTrACS v04r00 archive for all storms making landfall within 200 km of the study area between 1990 and 2024. Tracks are coloured by Saffir-Simpson equivalent intensity at landfall. The three treatment cyclones (Fani, Amphan, Yaas) are highlighted in bold. The spatial concentration of landfall events on the Odisha–West Bengal coast, and the May-window timing of the three treatment events, are visually apparent. Inset: kernel density estimate of Bay of Bengal cyclone landfall DOY from 1990 to 2024, showing the bimodal distribution (pre-monsoon May peak and post-monsoon October–November peak). [Source file: [figures/figS1_cyclone_climatology.pdf](#)]

Figure S2. Identification DAG: boundary-quantisation artefact pathway and the QC intercept. Pearl-style directed acyclic graph (DAG) showing the two parallel pathways linking a Bay of Bengal cyclone landfall to an observed shift in district-median Kharif rice EOS in a Sentinel-2 phenometric panel. Pathway 1 (agronomic, top): cyclone landfall → storm-surge inundation and rainfall anomaly → genuine transplanting or harvest delay → real phenometric shift → DiD coefficient. Pathway 2 (artefact, bottom): cyclone landfall → late-senescence pixels in coastal districts → NDVI trajectory does not complete within fitting window → optimiser assigns boundary DOY (349 for EOS, 288 for POS) → artefactual mode spike that is systematically more prevalent in treatment districts → spurious DiD coefficient. District and year fixed effects do not break the artefact arm because boundary contamination is correlated with treatment status (coastal districts have more late-harvest pixels). The three-gate QC operator (this paper) intercepts the artefact arm by removing all panel cells whose mode-share exceeds the 0.20 threshold or whose pixel-level fit-quality is below the RMSR < 0.15 cutoff. After QC, only the agronomic arm survives to the DiD estimator; the null result (Table 3) is therefore interpretable as a genuine agronomic null rather than as evidence offset by an opposing artefact. [Source file: [figures/figS2_identification_dag.pdf](#)]

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